relation between respitory diseases and air pollution

Jesper Smit 2894491,

23 June 2026

Set-up your environment

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(yaml)
library(rmarkdown)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```r
library(naniar)
library(tinytex)
library(ggplot2)

require(tidyverse)
require(rmarkdown)
require(yaml)
library(tidyr)
library(writexl)
library(readxl)
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```r
library(rnaturalearth)
library(rnaturalearthdata)
```

```
##
## Attaching package: 'rnaturalearthdata'
##
## The following object is masked from 'package:rnaturalearth':
##
##     countries110
```

```r
library(ggrepel)
library(countrycode)
```

Title Page

The relastion between respiratory diseases Include your names

Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social problem

1.1 Describe the Social Problem

Air pollution is the biggest current environmental health risk in the European Union, according to the European Environment Agency. The main risk of air pollution is respiratory disease, diseases like asthma or COPD. Each year, over 8 million people die from respiratory diseases (UT Health-San Antonio and South Texas Veterans Health Care System, 2021). In this paper, the relation between the severity of the air pollution and the frequency of respiratory disease is researched.

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```r
#loading the database
ghs <- read_csv("ghs.csv")
```

```
## Rows: 1000000 Columns: 22
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (7): Country, Disease Name, Disease Category, Age Group, Gender, Treatm...
## dbl (15): Year, Prevalence Rate (%), Incidence Rate (%), Mortality Rate (%),...
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
gaq <- read_csv("gaq.csv")
```

```
## Rows: 331920 Columns: 24
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (4): Country, State, City, AQI_Bucket
## dbl  (19): PM2.5 (ug/m3), PM10 (ug/m3), NO (ug/m3), NO2 (ug/m3), NOx (ppb), ...
## date  (1): Date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

```r
#head(dataset)
```

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```r
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```r
# making sure every name is consitent and correct.
ghs <- janitor::clean_names(ghs)
gaq <- janitor::clean_names(gaq)

#Using the correct data types per column for ghs and gaq.
ghs <-
  ghs |>
  mutate(country = as_factor(country),
         gender = as_factor(gender),
         disease_name = as_factor(disease_name),
         disease_category  = as_factor(disease_category),
         age_group  = as_factor(age_group),
         treatment_type  = as_factor(treatment_type),
         availability_of_vaccines_treatment  = as_factor(availability_of_vaccines_treatment)
  )
gaq <-
  gaq |>
  mutate(country = as_factor(country)
  )

#deleting other disease_category than respiratory for ghs
ghs <- ghs %>%
  filter(disease_category == "Respiratory")
#Only rows from 2014 or later for ghs
ghs <- ghs %>%
  filter(year >= 2014)
#Only rows till 2025.
ghs <- ghs %>%
  filter(year >= 2014)
#deleting rows with countries that are not in both datasets.
ghs <- ghs %>%
  filter(!country %in% c("Italy", "Canada", "Argentina"))
gaq <- gaq %>%
  filter(!country %in% c("Bangladesh", "Egypt", "Ethiopia", "Pakistan", "Phillipines", "Thailand", "Vie




#Deleting states and cities, by taking the avarage from them to only have countries.
gaq_country_year <- gaq %>%
  mutate(year = format(date, "%Y")) %>%
  group_by(country, year) %>%
  summarise(
    across(where(is.numeric), ~ mean(.x, na.rm = TRUE)),
    .groups = "drop"
  )
gaq <- gaq %>% select(-state)

#Deleting useless variables for gaq
gaq_country_year <- gaq_country_year %>%
  select(-c(wind_speed_km_h, humidity_percent, deforestation_rate_percent, aqi))
#Deleting useless variables for ghs.
ghs_country_year <- ghs %>%
```

```r
  select(-c(prevalence_rate_percent, age_group, gender, healthcare_access_percent, doctors_per_1000, hos

#Bringing all types off respiratory diseases in a year, to one year.
ghs_country_year <- ghs_country_year %>%
  group_by(country, year) %>%
  summarise(
    population_affected = sum(population_affected, na.rm = TRUE),
    incidence_rate_percent = mean(incidence_rate_percent, na.rm = TRUE),
    mortality_rate_percent = mean(mortality_rate_percent, na.rm = TRUE),
    recovery_rate_percent = mean(recovery_rate_percent, na.rm = TRUE),
    per_capita_income_usd = mean(per_capita_income_usd, na.rm = TRUE),
    education_index = mean(education_index, na.rm = TRUE),
    urbanization_rate_percent = mean(urbanization_rate_percent, na.rm = TRUE),
    .groups = "drop"
  )





#Making year the correct datatype for both datasets.
gaq_country_year <- gaq_country_year %>%
  mutate(year = as.numeric(year))

ghs_country_year <- ghs_country_year %>%
  mutate(year = as.numeric(year))


#Making the names of countries the same in both datasets
ghs_country_year$country <- countrycode(
  ghs_country_year$country,
  origin = "country.name",
  destination = "country.name"
)

gaq_country_year$country <- countrycode(
  gaq_country_year$country,
  origin = "country.name",
  destination = "country.name"
)

#Making the first variable. In generate necesarry date we explain all variables.
gaq_country_year <- gaq_country_year %>%
  mutate(
    total_air_pollution = rowMeans(scale(select(., pm2_5_ug_m3, pm10_ug_m3,
                                                no_ug_m3, no2_ug_m3,
                                                n_ox_ppb, nh3_ug_m3,
                                                co_mg_m3, so2_ug_m3,
                                                o3_ug_m3, benzene_ug_m3,
                                                toluene_ug_m3, xylene_ug_m3)),
                                   na.rm = TRUE)
  )
```

```r
#Selecting only the variables that we want to merge for the merged dataset
gaq_clean <- gaq_country_year %>%
  select(country, year, total_air_pollution)
#Merging the datasets
result <- gaq_clean %>%
  inner_join(ghs_country_year, by = c("country", "year"))

#Making variables 2 and 3. In generate necesarry date we explain all variables
result <- result %>%
  mutate(
    disease_intensity = scale(population_affected),
    disease_per_pollution = population_affected / (abs(total_air_pollution) + 1)
  )

#saving datasets
saveRDS(gaq_country_year, "gaq_country_year.rds")
saveRDS(ghs_country_year, "ghs_country_year.rds")
saveRDS(result, "result.rds")
```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

I already create the variable in data cleaning, because I use functions for data cleaning that contain that variable. So here I explain what they do; in data cleaning, the code is indicated with # to show which code it is.

Variable 1: Total air pollution is an index created by standardising the different pollutants and then summing them. Ultimately, an average is obtained that is compared to the pollution of all countries in the merged dataset.

Value >0: more polluted than average, value = 0: moderately polluted, value 0<: less polluted on average.

Variable 2 en 3 I already create the variable in data cleaning, because I use functions for data cleaning that contain that variable. So here I explain what they do; in data cleaning, the code is indicated with # to show which code it is.

Variable 2: Disease intensity shows whether people are affected by respiratory diseases more or less than average this year: 1 = much higher disease burden, 0 = average, -1 = much lower disease burden. It takes the average of all countries and rows.

Variable 3: Disease per population ratio, which shows how pollution compares to the number of diseases. Low value = fewer disease cases relative to pollution; high value = a lot of disease cases despite low/medium pollution.

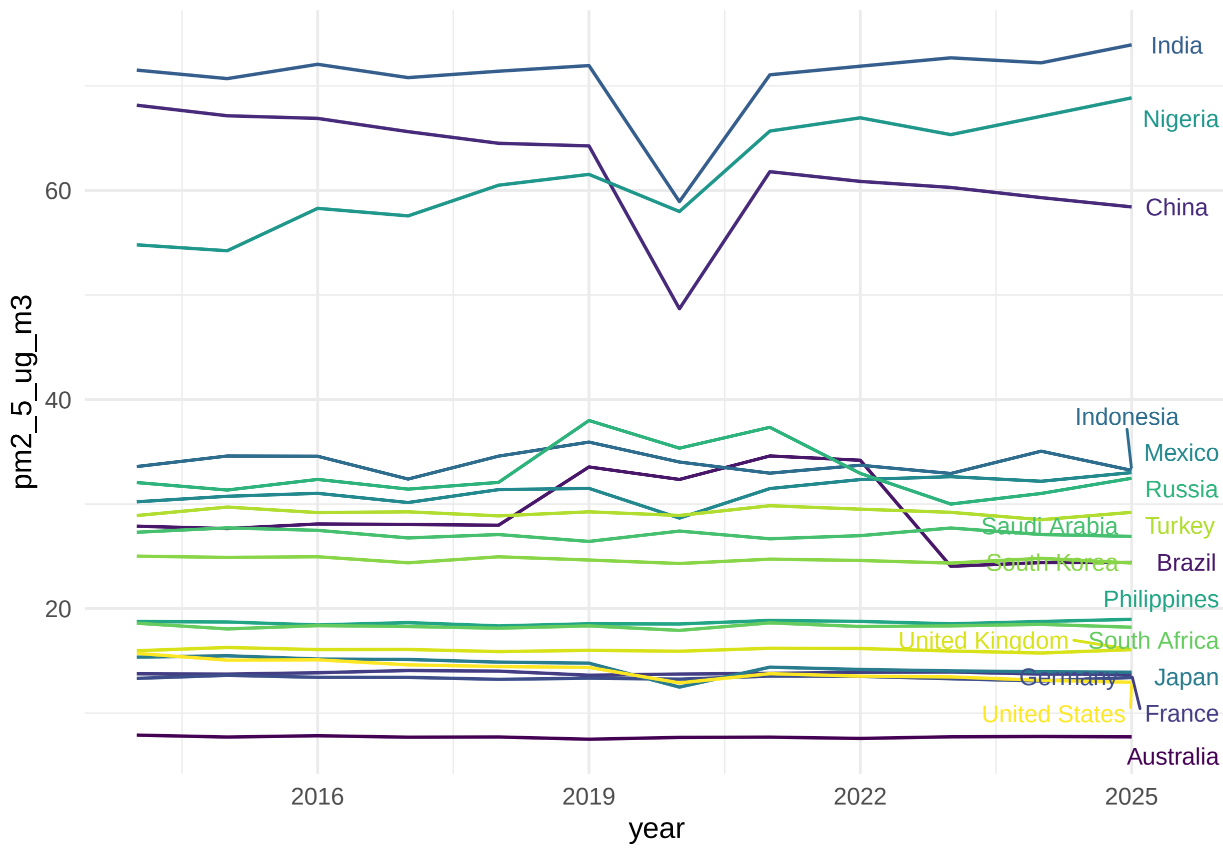
3.3 Visualize temporal variation

```r
library(dplyr)
library(tidyr)
library(ggplot2)

# data opschonen
gaq_country_year_clean <- gaq_country_year %>%
  filter(!is.na(pm2_5_ug_m3), !is.na(year))

# laatste punten per land
last_points <- gaq_country_year_clean %>%
  group_by(country) %>%
  filter(year == max(year)) %>%
  ungroup()

# plot
ggplot(gaq_country_year_clean,
       aes(x = year,
           y = pm2_5_ug_m3,
           color = country,
           group = country)) +
  geom_line(linewidth = 0.6) +
  geom_text_repel(data = last_points,
                  aes(label = country),
                  size = 3,
                  nudge_x = 0.5,
                  show.legend = FALSE) +
  scale_color_viridis_d() +
  theme_minimal() +
  theme(legend.position = "none")
```



3.4 Visualize spatial variation

library(dplyr) library(rnaturalearth) library(rnaturalearthdata) library(sf) library(plotly)

Data

data <- global_air_quality_2014_2025 names(data) <- make.names(names(data))

pm25_country <- data %>% group_by(Country) %>% summarise(PM2.5 = mean(PM2.5..ug.m3., na.rm = TRUE))

Wereldkaart

world <- ne_countries(scale = "medium", returnclass = "sf")

Merge

map_data <- world %>% left_join(pm25_country, by = c("name" = "Country"))

Alleen landen met data

map_data <- map_data %>% filter(!is.na(PM2.5))

Centroid per land

centroids <- st_centroid(map_data)

coords <- st_coordinates(centroids) centroids$lon < -coords[,1]centroids$lat <- coords[,2]

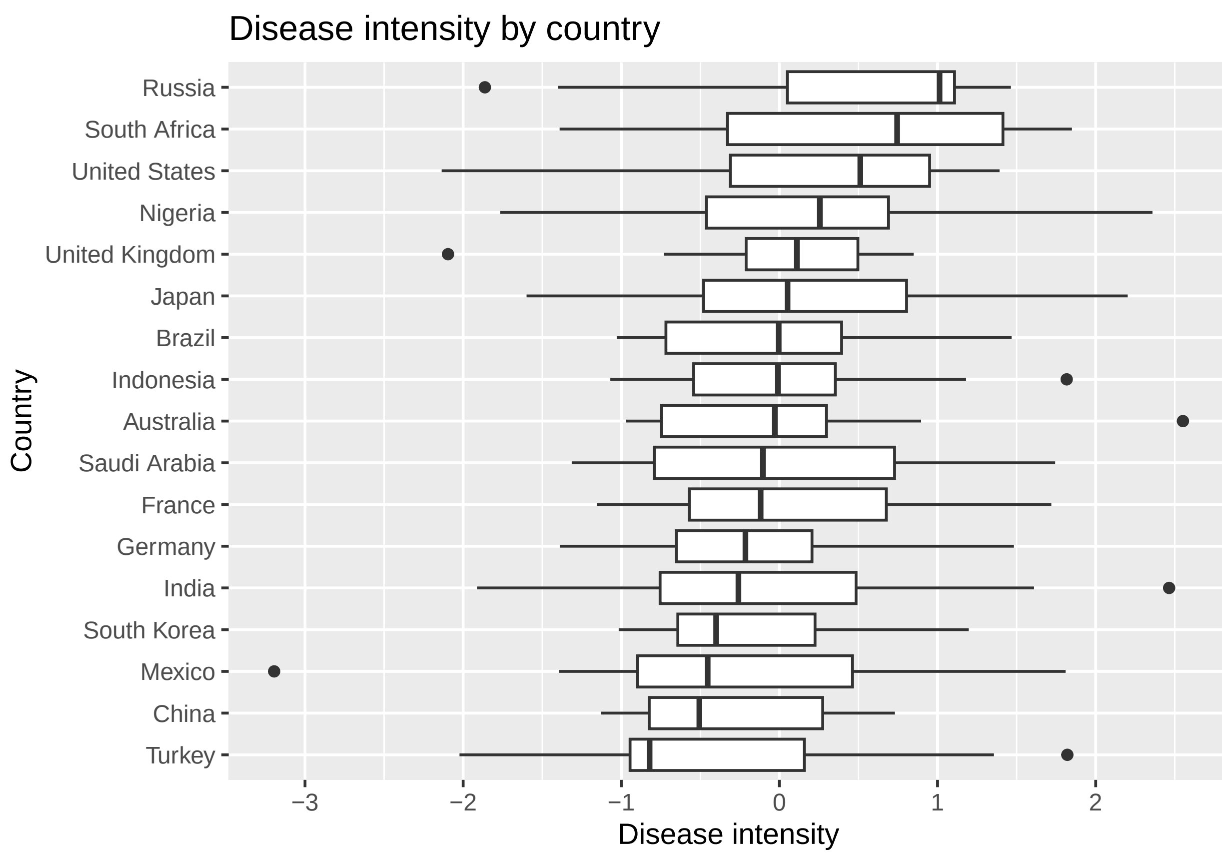
Plotly puntenkaart

fig <- plot_ly(data = centroids, lon = ~lon, lat = ~lat, type = "scattergeo", mode = "markers", marker = list(size = 6, color = ~PM2.5, colorscale = "Viridis", reversescale = FALSE, colorbar = list(title = "PM2.5")), text = ~paste("Land:", name, "PM2.5:", PM2.5), hoverinfo = "text") %>% layout(title = "Interactieve Wereldkaart PM2.5 (gemiddelde 2014–2025)", geo = list(projection = list(type = "natural earth"), showland = TRUE, landcolor = "rgb(230, 230, 230)", showcountries = TRUE, countrycolor = "rgb(200, 200, 200)"))

fig

3.5 Visualize sub-population variation

```r
#box plot
ggplot(result, aes(x = reorder(country, disease_intensity, median),
                   y = disease_intensity)) +
  geom_boxplot() +
  coord_flip() +
  labs(
    title = "Disease intensity by country",
    x = "Country",
    y = "Disease intensity"
  )
```



The boxplot reveals a high difference in disease intensity across countries. Russia and South Africa have a high median (the black lines in the boxes) compared with other countries; this suggests a higher disease burden than in other countries. Variability differs across countries, with some having a more unstable disease intensity (as seen by the size of the box). There are some outliers suggesting extreme spikes in disease intensity (indicated by the dots), which may be temporary shocks or faults in the measurements. The results show a high variability between countries.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: . . .

5.2 Reference list

European Environment Agency. (2024, January 25). Air pollution. Www.eea.europa.eu. https://www.eea.europa.eu/en/topics/in-depth/air-pollution

Levine, S., Marciniuk, D., Aglan, A., Celedón, J., Fong, K., Horsburgh, R., Malhotra, A., Masekela, R., Mortimer, K., Redde, H., Rice, M., & Simonds, A. (n.d.). UT Health-San Antonio and South Texas Veterans Health Care System. https://firsnet.org/wp-content/uploads/2025/01/FIRS_Master_09202021.pdf